

Mathematical Foundation and End-to-End Architecture of PRISM

1. Introduction

Credit scoring plays a fundamental role in modern lending by enabling financial institutions to estimate the probability that a borrower may default on future obligations. Traditional credit scoring systems primarily rely on bureau data and historical credit records, which often exclude new-to-credit and thin-file borrowers. To address this limitation, PRISM (Predictive Risk Insights and Scoring Model) incorporates alternative financial data sources including bank statements, salary slips, and utility payment records to generate explainable and robust credit risk assessments.

The mathematical foundation of PRISM is built upon established scorecard methodologies widely adopted across the banking industry. The framework integrates Feature Engineering, Optimal Binning, Weight of Evidence (WoE), Information Value (IV), Logistic Regression, Probability of Default (PD) estimation, Credit Score Scaling, SHAP-based explainability, Confidence Scoring, and Population Stability Monitoring. Recent research demonstrates that such interpretable scorecard-based approaches continue to offer strong predictive performance while maintaining regulatory compliance and transparency requirements.

2. PRISM System Architecture and End-to-End Scoring Logic

PRISM follows a structured multi-stage credit scoring pipeline designed to transform raw borrower documents into an explainable risk score.

Stage 1: Document Ingestion

Borrowers provide:

- Bank Statements
- Salary Slips
- Utility Bills

These documents undergo:

- OCR extraction

- Field parsing
- Data validation
- Format normalization

The objective is to convert unstructured financial documents into structured datasets suitable for risk analysis.

Stage 2: Feature Engineering

Financial behavior indicators are extracted from validated datasets, including:

- Credit-Debit Ratio
- Cashflow Variability
- Net-to-Gross Salary Ratio
- Utility Payment Stability
- Minimum Account Balance
- Salary Regularity

These variables form the quantitative representation of borrower behavior.

Stage 3: Optimal Binning

Continuous variables are converted into risk-based bins. Similar risk groups are clustered together while maximizing separation between good and bad borrowers.

Stage 4: Weight of Evidence Transformation

Each bin is transformed into a Weight of Evidence value representing the relative concentration of good and bad borrowers.

Stage 5: Logistic Regression

WoE-transformed features are fed into a logistic regression model which estimates borrower log-odds.

Stage 6: Probability of Default Estimation

Log-odds are converted into a Probability of Default (PD), representing the likelihood that a borrower may default.

Stage 7: Credit Score Scaling

PD values are transformed into an industry-standard credit score ranging from 300 to 900 using Points-to-Double-Odds (PDO) scaling.

Stage 8: Explainability Layer

SHAP-based explanations and reason codes identify the factors contributing to the final score.

Stage 9: Confidence Layer

Confidence scores quantify the reliability of the generated assessment based on data quality, document coverage, fraud confidence, and model certainty.

Stage 10: Fraud Detection and Risk Output

Fraud detection modules evaluate document authenticity, identity consistency, and income plausibility before producing the final risk classification.

The complete PRISM pipeline can therefore be summarized as shown in the figure 2.1:

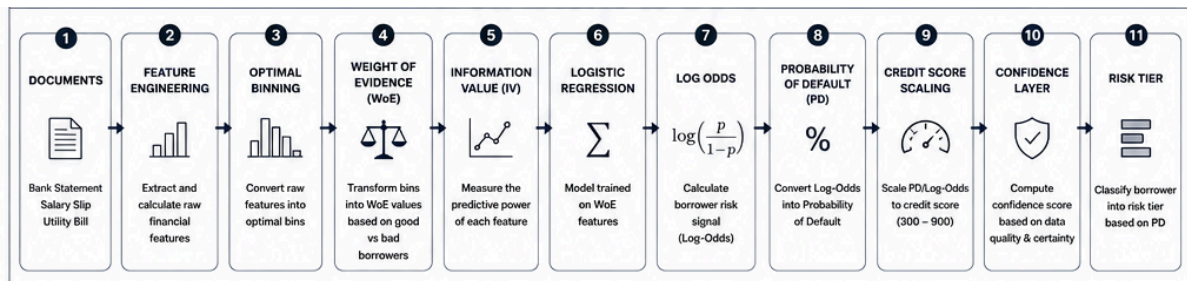


Fig 2.1 :- PRISM Credit Scoring Pipeline

3. Feature Engineering from Alternative Financial Data

Feature engineering represents the first mathematical stage of PRISM.

Research consistently shows that behavioral financial indicators provide strong predictive signals for borrower risk assessment.

3.1] Credit-Debit Ratio

Measures liquidity and spending behavior.

$$CDR = \frac{Total\ Credits}{Total\ Debits}$$

A higher ratio indicates stronger repayment capacity.

3.2] Cashflow Coefficient of Variation

Measures income consistency.

$$CV = \frac{\sigma}{\mu}$$

Where:

- (σ) = standard deviation of monthly cashflow
- (μ) = average monthly cashflow

Higher variability generally indicates elevated financial risk.

3.3] Net-to-Gross Salary Ratio

Measures salary quality and deduction burden.

$$NTG = \frac{\text{Net Salary}}{\text{Gross Salary}}$$

Higher values indicate healthier income retention.

3.4] Utility Stability Score

Measures consistency of recurring household expenses.

$$UtilityStability = \frac{\sigma_{bill}}{\mu_{bill}}$$

Stable utility behavior may indicate disciplined payment habits.

Recent studies have demonstrated that transactional banking behavior often outperforms traditional demographic variables in predicting borrower risk.

4. Weight of Evidence (WoE) and Information Value (IV)

Weight of Evidence remains one of the most widely adopted transformations in banking scorecards.

4.1] Weight of Evidence

$$WoE = \ln \left(\frac{\%Good}{\%Bad} \right)$$

Positive WoE values indicate lower-risk segments, while negative values indicate higher-risk segments.

Why WoE?

WoE encoding satisfies the linearity assumption of logistic regression by converting potentially non-linear relationships into monotonic risk signals, enabling stable coefficient estimation and improved interpretability.

This characteristic explains why WoE remains the industry standard in regulated credit scoring systems.

4.2] Information Value

Information Value quantifies the predictive strength of each feature.

$$IV = \Sigma (\%Good - \%Bad) \times WoE$$

Common interpretation:

IV Range	Predictive Strength
< 0.02	Not useful
0.02 – 0.1	Weak
0.1 – 0.3	Medium
0.3 – 0.5	Strong
> 0.5	Suspiciously predictive

PRISM's experimental results showed:

Feature	IV
Credit-Debit Ratio	0.82
Cashflow CV	0.77
Net-to-Gross Ratio	0.292
Utility Stability	0.087

These findings suggest that transactional banking features contribute most significantly toward borrower risk prediction.

5. Logistic Regression and Probability of Default Estimation

Following WoE transformation, the borrower risk profile is estimated using logistic regression.

5.1 Log-Odds Model

$$LogOdds = \beta_0 + \beta_1(WoE_{CDR}) + \beta_2(WoE_{Cv}) + \beta_3(WoE_{NTG}) + \dots + \beta_n(WoE_n)$$

Where:

- (β_0) = intercept
- (β_i) = learned model coefficients
- (WoE_i) = transformed risk features

The output represents borrower log-odds.

5.2] Probability of Default

Log-odds are converted into default probability using the logistic function.

$$PD = \frac{1}{1 + e^{-LogOdds}}$$

The resulting PD represents the estimated likelihood that a borrower may default.

The model achieved:

$$AUC = 0.817$$

An AUC of 0.817 indicates strong discriminatory power and approximately 81.7% ranking accuracy between risky and safe borrowers.

6. Credit Score Scaling Using PDO

While PD is useful mathematically, lenders typically require an intuitive score.

PRISM converts PD into a credit score between 300 and 900 using Points-to-Double-Odds (PDO) scaling.

$$\text{Factor :-} \quad \text{Factor} = \frac{PDO}{\ln(2)}$$

$$\text{Offset :-} \quad \text{Offset} = \text{BaseScore} - (\text{Factor} \times \ln(\text{BaseOdds}))$$

$$\text{Final Credit Score :-} \quad \text{CreditScore} = \text{Offset} + (\text{Factor} \times \ln(\text{BaseOdds}))$$

$$\text{Score:-} \quad \text{Score} = \text{BaseScore} + PDO \times \log_2(\text{Odds})$$

The final score is constrained within: $300 \leq \text{Score} \leq 900$

This methodology follows the industry-standard scorecard framework described by Siddiqi (2006).

7. Explainability Through SHAP-Based Reason Codes

Modern lending regulations increasingly require transparent decision-making.

PRISM incorporates SHAP-based explanations to identify how individual features influence the final score.

The SHAP value for a feature is computed as:-

$$\phi_i = \sum \frac{|S|! (M - |S| - 1)!}{M!} [f(S \cup \{i\}) - f(S)]$$

Where:

- (M) = total features
- (S) = feature subset
- (f(S)) = model prediction using subset (S)

SHAP provides:

- Local explanations
- Feature-level contributions
- Borrower-specific reason codes
- Regulatory transparency

This explainability layer allows PRISM to communicate why a score was assigned rather than simply reporting the final outcome.

8. Confidence Scoring Framework

Credit decisions should include an estimate of reliability.

PRISM computes confidence using four components:

$$Confidence = Coverage \times DocumentQuality \times FraudConfidence \times PredictionCertainty$$

Prediction certainty is defined as:

$$PredictionCertainty = |PD - 0.5| \times 2$$

Higher confidence values indicate greater trust in the generated assessment.

This framework allows the system to distinguish between:

- High-confidence decisions
 - Thin-file borrowers
 - Incomplete document submissions
 - Ambiguous risk profiles
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9. Fraud Detection and Identity Validation Framework

Traditional scorecards assume submitted information is genuine. PRISM introduces an additional fraud screening layer.

9.1] Income Anomaly Detection (FD-01)

Compares declared salary with actual bank credits.

$$IncomeAnomalyDetection = \frac{DeclaredIncome}{ObservedBankIncome}$$

Large deviations indicate potential income inflation.

9.2] Identity Consistency Checks (FD-02)

Cross-validates borrower identity across:

- Salary slips
- Bank statements
- Utility bills
- Government identifiers

Mismatch patterns trigger fraud flags.

9.3] Document Authenticity Validation (FD-03)

Checks:

- OCR consistency
- Missing metadata
- Template anomalies
- Duplicate document patterns

These mechanisms improve trustworthiness and reduce synthetic document fraud.

10. Population Stability and Model Monitoring

Credit scoring models must remain stable over time.

PRISM uses Population Stability Index (PSI) to monitor dataset drift.

$$PSI = \sum (Actual - Expected) \times \ln\left(\frac{Good}{Bad}\right)$$

Interpretation:

PSI	Meaning
< 0.10	Stable
0.10 – 0.25	Moderate Drift
> 0.25	Retraining Required

PRISM achieved:

$$PSI = 0.69$$

This indicates strong population stability and no immediate need for model recalibration.

11. Risk Tier Classification

Final borrower classification is derived from PD thresholds.

PD Range	Risk Tier
< 0.20	Low Risk
0.20 – 0.40	Medium Risk
0.40 – 0.70	High Risk
> 0.70	Very High Risk

This approach aligns risk interpretation directly with default probability estimates.

Conclusion

The literature demonstrates that PRISM is grounded in well-established credit risk methodologies while extending traditional scorecard models through alternative data integration, explainable AI, confidence estimation, and fraud detection mechanisms. The combination of WoE transformation, logistic regression, PD estimation, PDO score scaling, SHAP explainability, PSI monitoring, and fraud validation creates a transparent and regulator-friendly credit scoring framework. Experimental results showing an AUC of 0.817 and PSI of 0.069 further indicate that the model achieves strong predictive performance while maintaining stability across changing borrower populations.

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